



# Pyrolysis of mixed plastics for the recovery of useful products

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## ABSTRACT

Thermal and catalytic pyrolysis of polystyrene (PS) with low-density polyethylene (LDPE), high-density polyethylene (HDPE), polypropylene (PP), poly-ethylene terephthalate (PET) plastics were carried out in a 25 cm<sup>3</sup> stainless steel micro reactor at around 430–440 °C under 5.5–6.0 MPa of N<sub>2</sub> gas pressure for 1 h. Three reactions of each plastic with PS were conducted in the ratio of 1:1, 1:2 and 1:3. The amount of PS was varied to explore its role and reactivity. In all coprocessing reactions, ratio 1:1 afforded the best yields in the form of pyrolytic oils. SIM distillation of hexane soluble portion showed that the low boiling fractions were not found and fractions were obtained only after 96 °C + boiling point. It could be due to the vaporization of high volatile components. In most of the binary pyrolysis, light cycle oil (LCO) fractions have low recovery than heavy cycle oil (HCO). GC identified some very important chemical compounds present in the liquid products obtained from the pyrolysis of mixed plastics. The results obtained from this study have shown usefulness and feasibility of the pyrolysis process of the mixed plastics as an alternative approach to feedstock recycling.

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## 1. Introduction

The quantum growth of living standards and drastic increase in use of plastic products had a remarkable impact on the environment. Plastics have now become indispensable materials and the demand is continually increasing due to their diverse and attractive applications in household and industries. Mostly thermoplastics polymers make up a high proportion of waste and this amount is continuously increasing around the globe. Hence waste plastics pose a very serious environmental challenge because of their huge quantity and disposal problem as thermoplastics do not biodegrade for very long time. Plastics are produced from petroleum derivatives and are composed primarily of hydrocarbons but also contain additives such as antioxidants, colorants and other stabilizers. However, when plastic products are used and discarded, these additives are undesirable from environmental and pollution point of view [1–3]. Plastics are not presently biodegradable and are extremely troublesome components for land filling. Their destruction by incineration poses serious air pollution problems due to the release of harmful gases such as dioxins and hydrogen chloride, airborne particles and carbon dioxide [4].

Recycling has become a major response to the environmental challenges facing the plastic industry. Recycling has several meanings such as primary, secondary, tertiary and quaternary form of recycling. Tertiary recycling returns plastics to their constituent monomers or to their basic hydrocarbon feedstock [5,6]. The basic idea of tertiary

recycling is to break polymer molecules up into smaller organic molecules or monomers. The most important and widely used tertiary recycling process is pyrolysis, where plastics are heated in the absence of oxygen. Tertiary recycling of waste plastics may not be considered as a solution for waste plastics disposal only rather generating an environmentally acceptable transportation fuel or feed stock for the production of virgin plastics also [7,8]. Thermal degradation of polyolefins has been investigated extensively since Second World War [9–11]. There have also been reports on the pyrolysis of polyolefins to aromatic hydrocarbons with activated carbon-supported metal catalysts [12,13]; cracking of polystyrene and polyethylene on silica alumina [14] and reforming of heavy oil from waste plastics using zeolite catalysts [15,16] and others [17–19]. There has been a lot of research work on the pyrolysis of individual and mixed plastics to get liquid fuels [20–23].

Pyrolysis of polyethylene produces olefins of varying chain length and aromatic polymers produce aromatic compounds such as benzene, toluene and styrene. Pyrolysis can tolerate mixtures of different plastic resins as well as high levels of organic contamination, which is also pyrolyzed [24–26]. Western Research Institute has developed an advanced tertiary plastics recycling process that produces a product in the gasoline and diesel fuel boiling range for use in the refining and petrochemical industries [27]. It was also reported that the addition of some PS to polyolefin feedstock (PE, PP) accelerates the pyrolysis and leads to decomposition of products with a shorter chain length, e.g. more volatile products. More oil was obtained than expected from addition of both individual polymers [28]. Pyrolysis of electronic and thermoset were also investigated for the recovery of useful liquid products [29–31]. Korea Institute of

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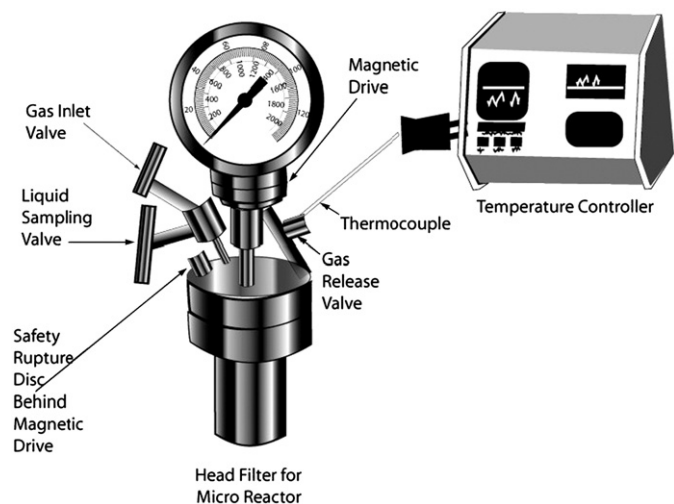


Fig. 1. Schematic experimental setup of micro-reactor under  $N_2$  gas pressure.

Energy Research has developed, demonstrated and described the process performance of pilot plants for the pyrolysis process of mixed plastic wastes generated from household and industry. The liquid products obtained by pyrolysis at around  $400\text{ }^\circ\text{C}$  were used as an alternative fuel oil [32]. Pyrolysis of a waste plastic mixture (HDPE, LDPE, PP, PS = 3:2:3:1) into a liquid product was also carried out in a stirred semi-batch reactor at low and high temperatures [33,34].

The current literature review indicates the importance of recycling of waste plastics around the globe. The present study is concerned with the investigation to develop a processing methodology suitable for the direct liquefaction of Saudi Arabia's consumer waste plastics into useful petrochemical liquid products. Saudi Arabia is one of the major producers of plastic in the world with total production capacity of around six million metric tons per year. The amount of plastic wastes in Saudi Arabia are about 15 wt.% in the composition of domestic municipality waste. Saudi Arabia is also one of the largest oil producing country in the world. The dominant components of waste plastics such as polyethylene, polystyrene, polypropylene are rich in carbon and hydrogen which are the building blocks of petroleum. Waste plastics have been pyrolyzed as individual plastic types and as mixed-plastics, there are little known about studies relating to the

interaction of plastic types in a mixture. LDPE, HDPE and PP plastics give a liquid oil product dominated by alkenes, alkanes, and alkadienes, and appreciable amounts of gases, and IOM. Pyrolysis of individual polystyrene produced lowest viscosity oil consisting of mainly styrene together with other aromatic hydrocarbons, lower yield of gases and no IOM formation. Therefore, polystyrene was chosen as the base plastic with which the other plastics (LDPE, HDPE, PP and PET) were mixed. Thermal and catalytic conversion of mixed plastic wastes into useful products has been achieved in this research project.

## 2. Experimental

### 2.1. Materials

#### 2.1.1. Polymers

The model plastic compounds, low density polyethylene (LDPE,  $M_n = 80,000$ ), high density polyethylene (HDPE,  $M_n = 85,000$ ), polyethylene terephthalate (PET,  $M_n = 32,500$ ), polypropylene (PP,  $M_n = 75,000$ ) and polystyrene (PS,  $M_n = 180,000$ ) used in this study were obtained from Saudi Basic Industries Corporation (SABIC), Riyadh, Saudi Arabia, and were used as received.

#### 2.1.2. Catalysts

A hydroprocessing catalyst available to us was used in this study. The hydroprocessing catalyst has following characteristic properties:  $MoO_3$  content = 13.9 to 17.0 wt.%, NiO content = 4.4 to 5.6 wt.%, Nominal Size = 1/16 in. cylindrical extrudate, Avg. Bulk Density = 44 lb/ft<sup>3</sup>, Loss on Ignition ( $500\text{ }^\circ\text{C}$ ) = 2.2 wt.%, Crush Strength = 12 lb, Avg. Diameter = 0.06 to 0.072 in. and Avg. Length = 0.09 in.

#### 2.1.3. Solvents and others

Locally available 99.9% pure  $N_2$  gas and HPLC grade hexane, toluene and tetrahydrofuran (THF) solvents from Fischer Scientific were used in the reactions.

### 2.2. Analytical procedure

Pyrolytic reactions were carried out in a 25 cm<sup>3</sup> stainless steel tubular micro autoclave reactor under different conditions of weight and type of catalysts, time duration, pressure and temperature with a programmable temperature controller obtained from the Parr

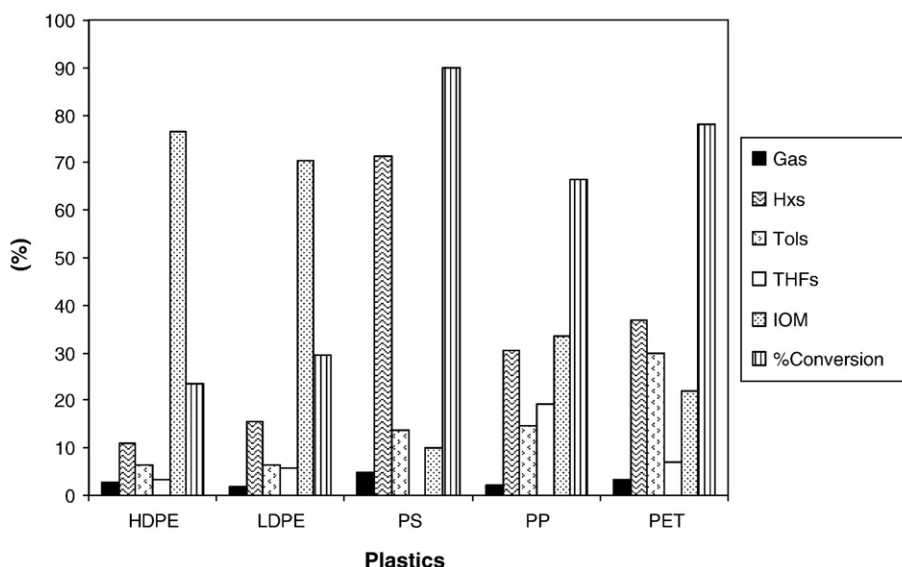


Fig. 2. Product distributions from the thermal pyrolysis of individual plastics.

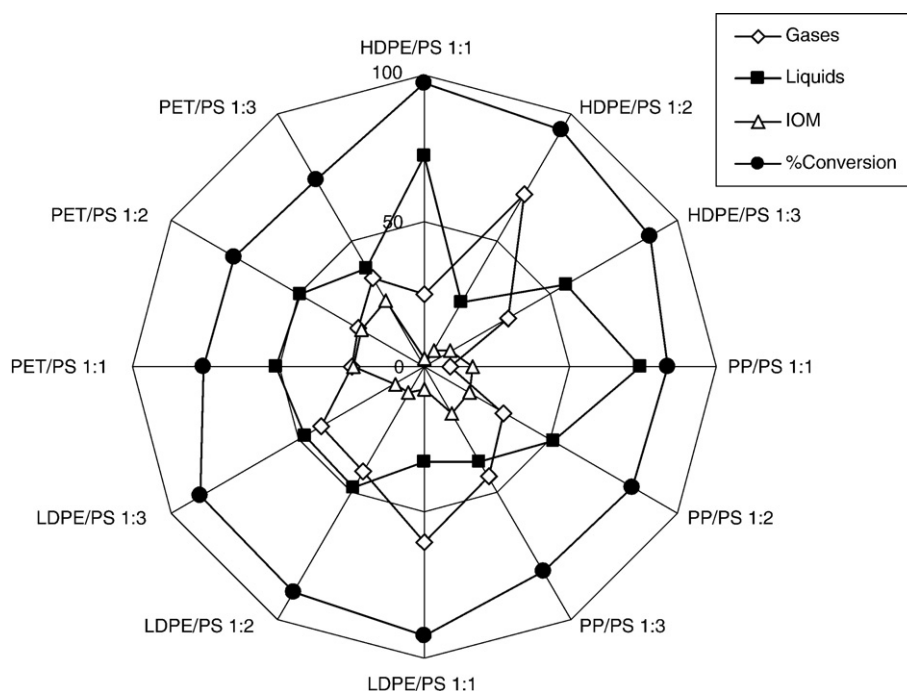


Fig. 3. Product yield from thermal pyrolysis of mixed plastics.

Scientific, USA. The operating conditions were evaluated for maximizing the desired liquid products. First, a series of thermal cracking of all plastics, HDPE, LDPE, PP, PS, and PET individually were carried out in a micro autoclave reactor under nitrogen atmosphere in order to establish their thermal behavior and reactivity. Then, in second step of the binary thermal reactions of individual HDPE, LDPE, PP, PET and waste plastic bottle were conducted with PS. Third, the binary catalytic reactions of individual HDPE, LDPE, PP and PET were carried

out with PS and catalyst to evaluate the reactivity and interactive effects among the reactants. The optimum conditions selected for our work were: 1% catalyst by weight of total feedstock weight, time 60 min, temperature between 430 and 440 °C under 5.5–6.0 MPa of  $N_2$  gas pressure. The micro reactor was agitated by a built-in mechanical stirrer and gradually heated with a temperature controlled built-in heater at the rate of 5 °C/min. The gas pressure throughout the experiments remained between 5.5 and 6.0 MPa at ambient

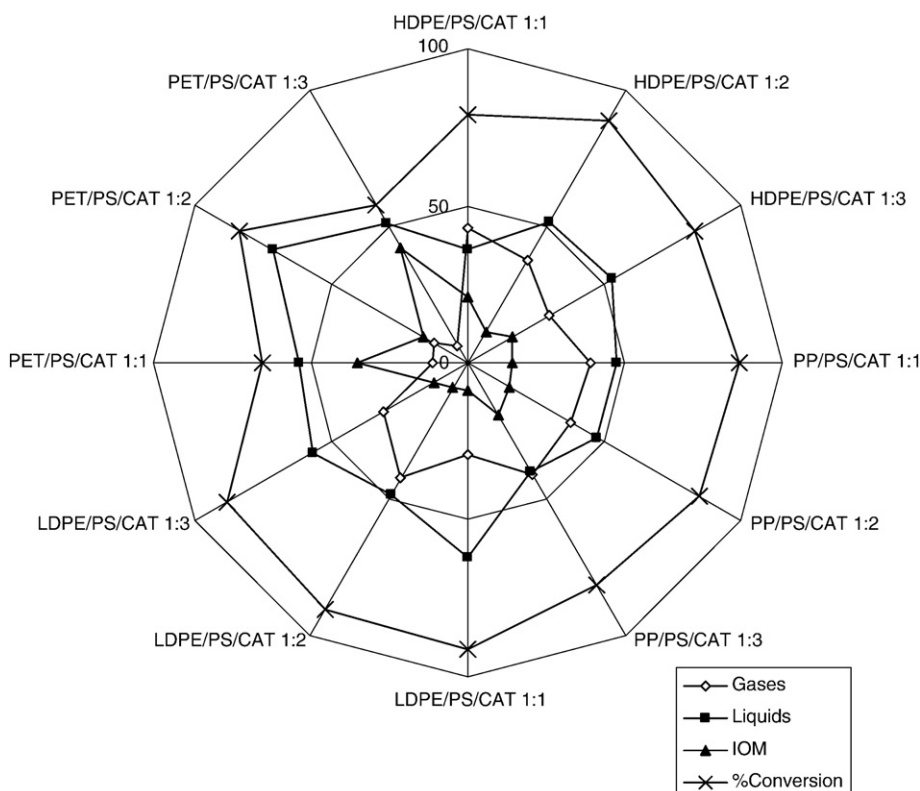


Fig. 4. Product yield from catalytic pyrolysis of mixed plastics.

**Table 1**  
Product yield (wt.%) from pyrolysis of three mixed plastics

| Plastics          | Gases <sup>a</sup> | Hxs  | Tols | THFs | IOM  | % Conversion |
|-------------------|--------------------|------|------|------|------|--------------|
| HDPE/PET/PS 1:1:2 | 45.4               | 19.3 | 15.8 | 2.6  | 17.0 | 83.0         |
| HDPE/PP/PS 1:1:2  | 49.5               | 35.9 | 3.5  | 0.9  | 10.3 | 89.8         |
| LDPE/PET/PS 1:1:2 | 27.5               | 31.5 | 18.4 | 6.1  | 16.5 | 83.5         |
| LDPE/PP/PS 1:1:2  | 49.5               | 32.4 | 1.8  | 0.9  | 15.5 | 84.5         |
| PET/PP/PS 1:1:2   | 53.3               | 25.4 | 2.6  | 1.8  | 17.0 | 83.0         |

<sup>a</sup>By difference.

temperature with nitrogen and evolved gases. After 1 h, the reactor was cooled to room temperature keeping under nitrogen and the gases evolved during the pyrolysis were vented out. Fig. 1 shows the schematic experimental setup of micro-reactor under N<sub>2</sub> gas. The pyrolysis of following mixed plastics was also conducted under similar reaction conditions: HDPE/PP/PS, HDPE/PET/PS, LDPE/PET/PS, LDPE/PP/PS and PET/PP/PS.

### 2.3. Separation and calculation of liquid products

After the reaction, the gaseous, liquid and solid products were separated and analyzed by the solvent fractionation methods to determine the amounts of liquid products using previously reported methods [8,21].

### 2.4. Fourier transform-infrared (FTIR) spectroscopy

The hexane solubles (HXs) obtained from different pyrolysis reactions were analyzed using infrared spectroscopy on a Perkin Elmer 16F PC FT-IR system. The thin film samples were prepared in chloroform. The samples were analyzed using thin film on the KBr salt plate technique. The insoluble organic materials (IOM) were analyzed using KBr as a mulling agent.

### 2.5. Nuclear magnetic resonance (NMR) spectroscopy

Solutions for the measurement of <sup>13</sup>C NMR spectra were prepared by dissolving 1 g of sample in 2 ml of deuteriated-chloroform containing 10 mg ferric acetyl acetonate, Fe(acac)<sub>3</sub>, as the relaxation agent. Solutions for the measurement of <sup>1</sup>H NMR spectra were prepared, in a similar way, as described for <sup>13</sup>C spectra with the exception that no relaxation reagent was used. 0.5 ml of liquid products was dissolved in 2 ml of deuteriated-chloroform. <sup>1</sup>H NMR spectra were recorded on a JEOL-500 pulse Fourier transform (PFT) spectrometer operating at 500 MHz using 5 mm sample tubes. <sup>1</sup>H NMR spectra were obtained and the experimental conditions were: spectrum width, 2800 Hz; data points, 15,680; pulse width, 3.5 ms (45°); pulse delay, zero; and number of transients, 64.

### 2.6. Simulated distillation

The boiling range distributions of hexane soluble fractions were determined using simulated distillation analyzer [ASTM method 5307]. The simulated distillation analyzer used for this study was a Hewlett Packard 5880A gas chromatograph equipped with automatic sampler model HP 7673A and a flame ionization detector (FID). The column used was 1/8×20 in. stainless steel, coated with 10% UCW-982 on 80/100 mesh chromosorb P-AW. The computer program calculates the data and produces a simulated distillation report in percent distilled versus temperature.

### 2.7. Gas chromatography

Agilent GC 6890 plus Series gas chromatograph equipped with a split-splitless injector (split ratios of 20:1) was used for the GC analysis. The temperature of the injector was 250 °C, with 10 psi

constant pressure. The column was an HP-5 column (30 m×0.25 mm i.d., 0.25 µm film thickness) consisting of Crossbond® (5% phenyl–95% dimethylpolysiloxane). Helium was the carrier gas at a flow rate of 15.8 ml/min and programmed temperature was applied to obtain the separation of the compounds; precisely the initial temperature was 50 °C, hold 2 min, then ramped at 10 °C/min to 140 °C (hold time 0 min), then finally ramped at 20 °C/min to 250 °C (hold time 20 min). The detector was flame ionization detector (FID), with hydrogen and air flow of 40.0 and 450.0 ml/min respectively. Makeup flow was on with 45.0 ml/min of helium.

### 2.8. Gas chromatography–mass spectrometry

A Trace GC-MS 2000 Series (Varian inc. USA) gas chromatograph equipped with a split-splitless injector (split ratios of 20:1) was used for the GC-MS analysis. The column was a Cp-Sil low bleed®-5 MS column (30 m×0.25 mm i.d., 0.25 µm film thickness) consisting of Crossbond®. Helium was the carrier gas at a flow rate of 1.0 ml/min and programmed temperature was applied to obtain the separation of the compounds; precisely the initial temperature was 50 °C, hold 2.0 min, then ramped at 10 °C/min to 140 °C (hold time 0 min), then finally ramped at 20 °C/min to 250 °C (hold time 20 min). The gas chromatograph was interfaced with a GCQ Plus mass detector operating in the EI mode (70 eV), under autotune conditions. The temperatures of injector, transfer line and ionization source were 250, 225 and 270 °C, respectively. The mass spectra were recorded within 50–450 amu in full scan mode to collect the total ion current (TIC) chromatograms; NIST MS search was used for compound identification.

## 3. Results and discussion

### 3.1. General properties and product yields

Some common properties and general uses of various plastics have been discussed earlier [8,21]. Each of the following polymers, LDPE, HDPE, PP, PET and PS, used in this study was pyrolyzed individually at optimal reaction conditions obtained. In order to establish the optimum reaction conditions, the pyrolysis reactions of thermal and catalytic systems were performed at different temperatures (300, 400, 430 and 500 °C), nitrogen gas pressure (4.0, 5.0 and 6.0 MPa) and time duration (40, 60 and 90 min). The best results, based on the percent

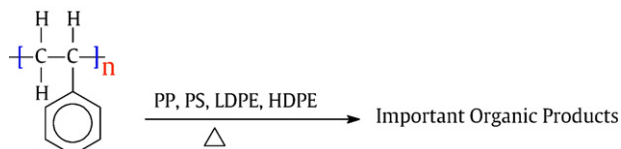
**Table 2**

Chemical compounds identified by GC in the hexane soluble fraction obtained from PET/PP/PS pyrolysis

| Peak # | RetTime [min] | Area % | Name of compounds                         |
|--------|---------------|--------|-------------------------------------------|
| 1      | 11.7          | 2.5    | Naphthalene                               |
| 2      | 12.0          | 7.2    | Benzoic acid                              |
| 3      | 13.2          | 1.6    | 2-Methyl Naphthalene                      |
| 4      | 13.4          | 1.4    | 1-Methyl Naphthalene                      |
| 5      | 14.2          | 6.8    | Biphenyl                                  |
| 6      | 14.6          | 1.5    | 1,7-Dimethyl Naphthalene                  |
| 7      | 14.7          | 5.5    | Diphenylmethane                           |
| 8      | 15.2          | 4.0    | 1,1'-Biphenyl-3-Methyl                    |
| 9      | 15.4          | 4.5    | 1,1'-Biphenyl-4-Phenyl                    |
| 10     | 15.6          | 2.4    | Bibenzyl                                  |
| 11     | 16.2          | 2.9    | 4-Ethylbiphenyl                           |
| 12     | 16.3          | 2.8    | 2,2'-Dimethyl Biphenyl                    |
| 13     | 16.9          | 1.9    | Naphthalene 2-Phenyl                      |
| 14     | 17.1          | 1.4    | Naphthalene 2 (Phenylmethyl)              |
| 15     | 18.5          | 1.9    | p-Ter Phenyl                              |
| 16     | 19.4          | 7.1    | m-Ter Phenyl                              |
| 17     | 21.3          | 19.1   | Butybenzyl Phthalate 1,1',3',1"-Terphenyl |
| 18     | 23.9          | 6.2    | Methanone, (1,1'-Biphenyl)-4-YLPhenyl     |
| 19     | 25.0          | 8.2    | 1,1',3',1"-Terphenyl-5'-Phenyl            |
| 20     | 26.2          | 4.7    | 1,1,3,1,1'-Quaterphenyl                   |
| 21     | 27.8          | 2.0    | Quaterphenyl                              |

conversion and amount of hexane soluble in our study, were obtained 1% catalyst by weight of total feedstock weight, time 60 min, pressure 5.5–6.0 MPa and temperature 430–440 °C. These optimum conditions were used for further studies.

The product distributions from single component thermal pyrolysis are shown in Fig. 2. The conversion for the single component reactions showed the nearly complete conversion was observed for PS whereas LDPE and HDPE yielded lower conversions and PP and PET remained in the moderate to high conversion. Polystyrene was degraded almost completely under normal thermal pyrolysis conditions in comparison to other polymers. It prompted us to explore the possibility of using polystyrene in the co-processing or pyrolysis reactions of mixed waste plastics.



Therefore, thermal and catalytic reactions containing binary combinations of the PS and LDPE, HDPE, PP, PET plastics were studied. The product distributions from the thermal coprocessing of these plastics and PS are shown in Fig. 3. Three reactions of each plastic with PS were conducted in the ratio of 1:1, 1:2 and 1:3. The amount of PS was varied to understand its reactivity. The result indicated that ratio 1:1 afforded best results in the form of conversion as shown in Fig. 3. This includes conversion of total charge (plastics + PS) into gases and liquids products that is called pyrolysis oils. Increasing the amount of

PS decreased the conversion percent. It shows that reactivity of PS is not open ended rather depends on the amount of feed. Since PS contains small aliphatic, non-polar and large aromatic, polar groups that might be hindering the conversion of other alkyl groups present in the plastics into liquids.

Fig. 4 shows the product distributions from catalytic coprocessing of plastics with PS. In these experiments, 1% catalyst by weight of total feedstock weight was used. The results indicate that amount of hexane solubles remain highest with 1:1 ratio of plastics and PS. However the pyrolysis of mixed waste plastics provides some different results as shown in Table 1. Pyrolysis of three plastics mixture afforded a maximum of 90% conversion and lowest was 83%. It shows that there is a definite reactivity among the plastics during the pyrolysis and more work is needed to investigate the exact nature of reaction processes taking place.

### 3.2. Gas chromatography and mass spectrometry analysis

Gas chromatography analysis of all liquid products was carried out to identify the types of components present. Table 2 shows most of the compounds identified using gas chromatography with flame ionization and mass spectrometric detection in liquid products obtained from the various pyrolysis processes. The liquid products gave almost exclusively substituted aromatic products without any alkanes, alkenes and alkadienes. A wide spectrum of hydrocarbon fragments which may contain any number of carbon atoms may be expected from the thermal degradation of the polyalkene mixed plastics, which occurs through random scission. The C–C bond is the weakest in the polyalkene structure and during the degradation process, the stabilization of the resultant radical after chain scission leads to the formation of carbon double bonds (C=C) in the structure.

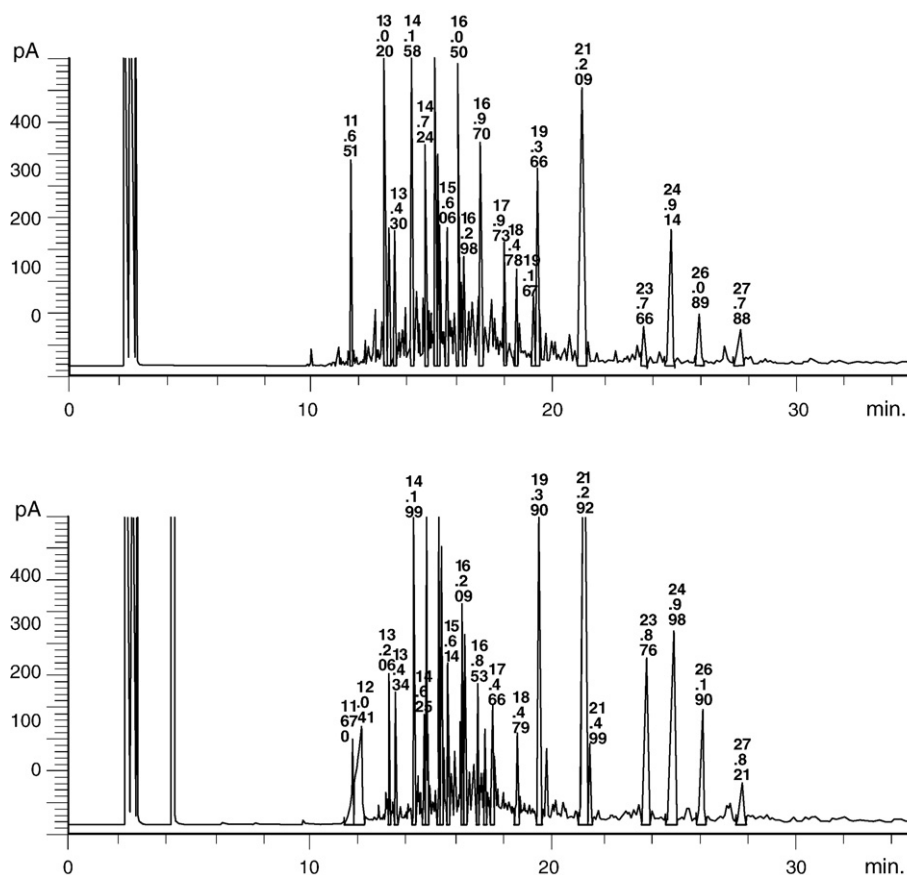
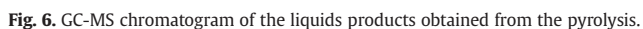


Fig. 5. Gas chromatogram of the liquids products obtained from the pyrolysis.



Mass spectra of the liquid product obtained from the catalytic pyrolysis of mixed plastics gave very important organic compounds. Some of the compounds searched through the GC-MS library identified: biphenyl, Diphenylmethane, Bibenzyl, 4-ethylbiphenyl, 1,1-diphenyl-2-ethyl, benzene-1-1-(1-methyl-1,3-propanediyl)bis, phenanthrene, 1-phenyl naphthalene, m-terphenyl, 2-phenylnaphthalene, 2-(Phenylmethyl)-naphthalene, p-terphenyl, and many more. Fig. 5 shows the GC and Fig. 6 shows the GC-MS chromatogram of the liquids products obtained from the pyrolysis.

The following prominent peaks were observed in FT-IR spectra of the liquid products obtained from various pyrolysis processes.

| No.        | Mixed plastics | % Aliphatic hydrogens | % Aromatic hydrogens |
|------------|----------------|-----------------------|----------------------|
| 1          | PP-PS          | 29.58                 | 70.42                |
| 2          | LDPE-PS        | 34.21                 | 65.79                |
| 3          | HDPE-PS        | 43.50                 | 56.50                |
| 4          | PET-PS         | 17.36                 | 82.64                |
| 5          | PP-PS-CAT      | 37.11                 | 62.89                |
| 6          | LDPE-PS-CAT    | 50.00                 | 50.00                |
| 7          | HDPE-PS-CAT    | 44.75                 | 55.25                |
| 8          | PET-PS-CAT     | 21.26                 | 78.74                |
| 9          | HDPE-PET-PS    | 27.54                 | 72.46                |
| 10         | HDPE-PP-PS     | 55.36                 | 44.64                |
| 11         | LDPE-PP-PS     | 39.76                 | 60.24                |
| 12         | PET-PP-PS      | 31.03                 | 68.97                |
| 13         | LDPE-PET-PS    | 40.12                 | 59.88                |
| Carbon NMR |                | % Aliphatic carbons   | % Aromatic carbons   |
| 1          | PP-PS          | 14.4                  | 85.6                 |
| 2          | PP-PS-CAT      | 24.8                  | 75.2                 |
| 3          | HDPE-PET-PS    | 21.9                  | 78.1                 |

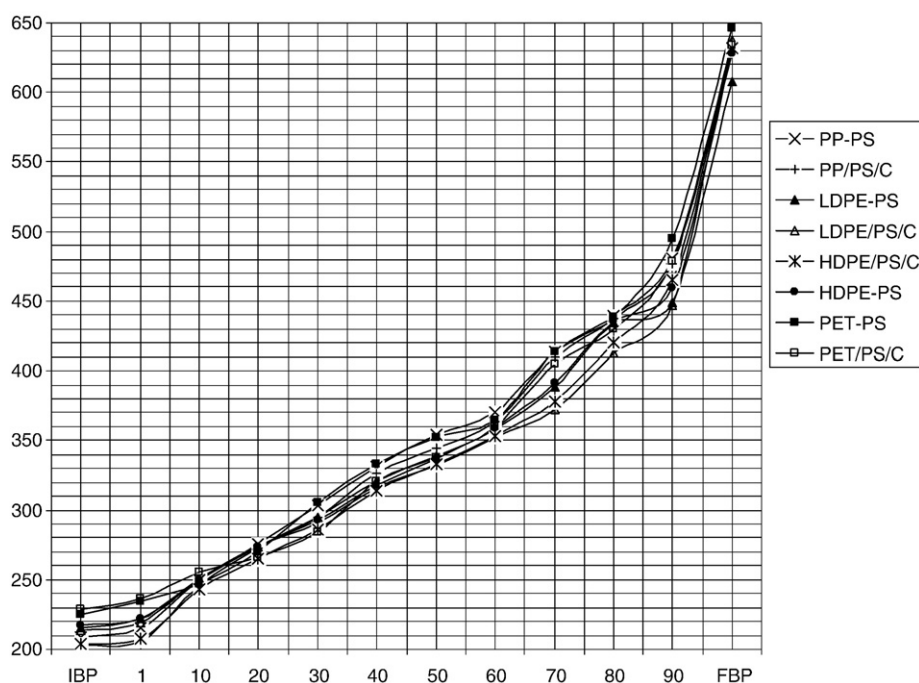


Fig. 7. Products distribution obtained from the SIM distillation of hexane solubles fractions of mixed plastic pyrolysis.

3084 (sharp and medium) and 3025 (sharp and strong) due to alkene C–H stretch

2919 (strong and sharp) asymmetrical –CH<sub>2</sub> stretch

2870 (broad and medium) symmetrical –CH<sub>2</sub> stretch

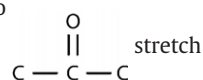
2733 (sharp and weak) due to carbonyl overtones

1941, 1857, 1801 (sharp and weak) due to overtones

1729 (sharp and medium) due to carbonyl

1603, 1494, 1460 due to C=C stretch

1178 (sharp and weak) due to



1080 and 1029 (sharp and medium) due to aromatic in-plane bending

894 (weak and sharp) due to C–H out-of-plane bend

699 and 730 (strong and sharp) due to C–H aromatic bending

The overall FT-IR spectra of the liquid product obtained from the various pyrolysis processes is, therefore, dominated by the presence of alkane and alkene functional groups. The presence of mono, polycyclic and substituted aromatic groups were clearly observed at lower wavelengths. Benzene has been identified as a major constituent in oils derived from the pyrolysis of PS, while other aromatic compounds identified included alkylated benzenes and naphthalene and other polycyclic aromatic hydrocarbons. The liquid products obtained from the pyrolysis of HDPE, LDPE and PP with PS gave very similar FT-IR spectra, which consist essentially of the mixed spectra from each component. Each of spectra contains the dominantly alkane and alkene peaks derived from the HDPE and LDPE plastics and the mainly aromatic peaks derived from the PS. The aromatic structures present in the oil from the mixtures are clearly indicated by the strong peaks from 3000 to 3050 cm<sup>−1</sup> and group of peaks between 675 and 900 cm<sup>−1</sup>.

### 3.4. NMR spectroscopy analyses

The detailed NMR analyses of the liquid products derived from the pyrolysis of mixed plastics are given in Table 3. The thermal

degradation of mixed plastics produces liquid that contains major portion of aromatic hydrogens. The catalytic pyrolysis afforded almost same results with not much significant changes in the hydrogen contents of the liquid products. PET/PS gave more aromatic hydrogen containing liquid product than HDPE and LDPE with PS. The higher aromatic content of hydrogen indicates the extent of interaction between polystyrene and other plastics. PP has an additional CH<sub>3</sub> group than HDPE and LDPE polymers so it was expected to give lower content of aromatic hydrogen but the results shown were in the contrary. Since thermal degradations are free radical initialized with random scission of bonds and that could have produced the CH<sub>3</sub> radicals which might have undergone further defragmentation. These CH<sub>3</sub> radicals might have affected and assisted in breaking up of other C–C bonds but these steps need to be further studied and experimentally verified.

The H NMR results obtained from the pyrolysis of three mixed plastics were also analyzed. Each of H NMR spectra contains the marginally dominant aromatic peaks derived from the thermal pyrolysis of two of these (LDPE, HDPE, PP and PET) plastics and PS. The integrated area of H NMR spectra illustrates the presence of –CH<sub>3</sub>, –CH<sub>2</sub> and C–H functional groups in the aliphatic region between 0:00 and 4.00 ppm. The aromatic structures present in the liquid portion obtained from the pyrolysis of mixed plastics clearly indicated by the strong peaks C=C absorbance in the between 6:00 and 8.00 ppm region. Most of the mixed plastics have shown lower content of aromatic hydrogen in comparison to mixing of two plastics. It was noticed that the pyrolysis of any two plastics and one of them PET mixes well with PS during thermal pyrolysis. The content of

Table 4

Thermal properties of individual model plastics

| Model and plastics waste | Crystalline melt temperature °C (T <sub>m</sub> /T <sub>g</sub> ) | On set temperature °C | % Crystallinity |
|--------------------------|-------------------------------------------------------------------|-----------------------|-----------------|
| LDPE                     | 118                                                               | 106.3                 | 28.9            |
| HDPE                     | 129.7                                                             | 123.1                 | 56.8            |
| PS                       | 230/104.4                                                         | 96.5                  | –               |
| PP                       | 164                                                               | 153.8                 | 28.1            |
| PET                      | 248.5                                                             | 240.8                 | –               |

aromatic hydrogens was found to be higher. It is clear from the result that the mixing or interaction of two plastics with PS has not very well taken place where as the mixing of any individual plastic with PS occurred in thermal as well as catalytic pyrolysis.

The carbon NMR results obtained from the pyrolysis of plastics were also analyzed. Each of carbon NMR spectra contained the dominant aromatic peaks. The results are complimentary and supporting the findings of H NMR results. The aromatic carbon contents were found to be higher in all liquid products obtained from the pyrolysis processes as shown in Table 3. Some interesting observations were made when PP-PS pyrolyzed thermally and catalytically. The aromatic carbon contents decreased significantly during the catalytic pyrolysis than thermal. These results are in line with the results obtained by the H NMR of both samples. It indicates more interaction of PS with PP so the aromatic ring of PS is either broken up or isomerizes. Thermal pyrolysis of HDPE-PET-PS afforded a moderate to high contents of aromatic carbons.

### 3.5. SIM distillation of liquid products

The hexane soluble fractions of the different reaction systems were analyzed using simulated distillation (SIM). The results from the analysis regarding the recovery temperatures of HXs at initial boiling point (IBP) show 10%, 50%, 90% and final boiling point (FBP). In addition, the amount of LCCG (light catalytically cracked gasoline, light naphtha), HCCG (heavy catalytically cracked gasoline, light naphtha), LCO (light cycle oil) and HCO (heavy cycle oil) were also determined. The SIM results show a general decrease in IBP recovery temperature for the liquid products. The low boiling fraction materials were not found and cuts were obtained only after 96 °C + boiling point. It could be due to the vaporization of highly volatile components. In most of the binary system, LCO fractions have low recovery than HCO fractions. Percent product distribution of different hydrocarbons obtained from the SIM distillation of hexane soluble portion is given in Fig. 7. The proportion of material that boiled in the range 200–300 and more than 300 °C was found to be relatively larger.

### 3.6. Thermal analysis

Thermal analysis was defined as a group of methods based on the determination of changes in chemical or physical properties of material as a function of temperature in a controlled atmosphere. Thermal analysis studies were carried out on model plastics including LDPE, HDPE, PP, PS and PET. Table 4 presents the crystalline melting temperature for crystalline polymers and glass transition temperature for crystalline/amorphous polymer present in the model plastics. The percent crystallinity indicates the crystallites presence that can also be a result of cross linking. The higher values of T<sub>m</sub> and T<sub>g</sub> for PS are the basis of pyrolysis reactions.

## 4. Conclusions

The thermal and catalytic pyrolysis of low-density polyethylene (LDPE), high-density polyethylene (HDPE), polypropylene (PP), polyethylene terephthalate (PET) and polystyrene (PS) plastics were studied individually and in mixed ratios with PS. The pyrolysis reactions were carried out in a 25 cm<sup>3</sup> stainless steel autoclave micro tubular reactor at around 430–440 °C under 5.5–6.0 MPa of N<sub>2</sub> gas pressure for 1 h. The result indicated that ratio 1:1 afforded best results in the form of conversion. In all catalytic coprocessing of mixed plastics, ratio 1:1 afforded the best yields in the form pyrolytic oils. The SIM distillation of hexane soluble portion showed that the low boiling fractions were not found and fractions were obtained only after 96 °C + boiling point. It could be due to the vaporization of high volatile components. In most of the binary pyrolysis, light cycle oil (LCO) fractions have low recovery than heavy cycle oil (HCO). GC

identified some very important chemical compounds present in the liquid products obtained from the pyrolysis of mixed plastics. These compounds are of great importance to the industry and environment as well. Thermal and catalytic pyrolysis of plastics with PS seem to be a feasible process by which plastics and PS can be converted into useful liquid fuels and chemical feed stocks. The results obtained from this study show usefulness and feasibility of the pyrolysis process of the mixed plastic as an alternative approach to feedstock recycling.

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